

Machine Learning Potentials

Lecture 11 — PHYG004, 2026 Spring

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Concepts and a guided tour of the key papers

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Outline

Part 1 — Why ML potentials?

- The accuracy–cost gap between DFT and classical force fields
- What we want from an interatomic potential

Part 2 — Core concepts

- Potential energy surface, locality, atomic decomposition
- Required symmetries ($E(3)$ + permutation) and how to enforce them
- Descriptor-based vs end-to-end (graph) approaches

Part 3 — Paper tour

- Behler–Parrinello (2007), GAP/SOAP, ANI, DeepMD
- SchNet, DimeNet, NequIP, MACE, Allegro

Part 4 — Foundation models, applications, open problems

Part 1

Why Machine Learning Potentials?

The accuracy–cost gap

Ab initio (DFT, CCSD(T), QMC)

- Predictive accuracy on bonding, reactions, defects
- Cost scales steeply: DFT $\mathcal{O}(N_e^3)$, CCSD(T) $\mathcal{O}(N_e^7)$
- Practical limit: $\sim 10^2$ – 10^3 atoms, $\sim$ ps timescales

Classical force fields (LJ, EAM, ReaxFF, ...)

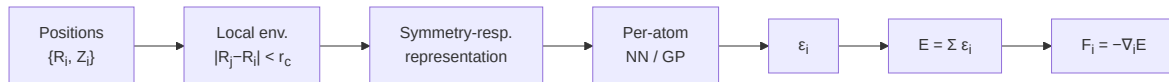
- Hand-crafted analytical forms, fitted to a few systems
- Cheap: 10^6 atoms $\times$ ns– μ s are routine
- Poor transferability; struggle with bond breaking, charge transfer

Many problems live in a gap that *neither* method covers: nucleation, phase transitions in complex alloys, defect dynamics, large-scale heterostructures, finite-T phonons, reactive interfaces.

What we want from an interatomic potential

Given N atoms with positions $\{\mathbf{R}_i\}$ and species $\{Z_i\}$, the potential is a scalar function with forces obtained by autodiff:

$$E(\{\mathbf{R}_i, Z_i\}) \in \mathbb{R}, \quad \mathbf{F}_i = -\nabla_{\mathbf{R}_i} E$$



Desired properties: **accurate** (DFT-level $E, \mathbf{F}, \sigma$), **smooth & differentiable**, **size-extensive**, **symmetry-respecting** ($O(3)$ + permutation), **transferable**, **cheap** ($\gg$ DFT).

A timeline of ML potentials

Year	Method	Key idea
2007	Behler–Parrinello (NNP)	Per-atom NN on hand-crafted symmetry functions
2010	GAP / SOAP (Bartók)	Gaussian processes + invariant power-spectrum descriptor
2017	ANI-1	Modified Behler–Parrinello, large organic dataset
2017	SchNet	First widely-used continuous-filter graph NN potential
2018	DeepMD	End-to-end learned descriptor + per-atom MLP
2020	DimeNet / DimeNet++	Directional (angle-aware) message passing
2022	NequIP	$E(3)$ -equivariant tensor-product message passing
2022	MACE	Higher-body equivariant messages via ACE
2022	M3GNet, CHGNet	First <i>universal</i> MLPs over the periodic table
2023	Allegro	Strictly local equivariant model, MD-scale parallel
2024	MACE-MP-0, MatterSim, ORB, SevenNet	Foundation models pretrained on MP / Alexandria

Part 2

Core concepts

The potential energy surface (PES)

The **Born–Oppenheimer PES** is the ground-state electronic energy as a function of nuclear coordinates:

$$E(\mathbf{R}_1, \dots, \mathbf{R}_N) = \min_{\Psi} \langle \Psi | \hat{H}_{\text{el}}(\{\mathbf{R}_i\}) | \Psi \rangle$$

This is the object we want to learn.

The PES is:

- a smooth function of nuclear positions almost everywhere
- a $3N$ -dimensional surface — too large to tabulate directly
- *not* uniquely defined per atom; the per-atom split we will use is a **modeling choice**, not a physical decomposition

Key question: how do we represent $E(\mathbf{R}_1, \dots, \mathbf{R}_N)$ as a finite-parameter function we can fit and that scales gracefully to $N \rightarrow \infty$?

Locality and the atomic energy decomposition

The **near-sightedness** of electronic matter (Kohn 1996, Prodan–Kohn 2005): a local change in the density mostly affects nearby physics. This motivates the universal modeling assumption behind *almost all* MLPs:

$$E(\mathbf{R}_1, \dots, \mathbf{R}_N) = \sum_{i=1}^N \varepsilon_i(\mathcal{N}_i), \quad \mathcal{N}_i = \{(\mathbf{R}_j, Z_j) : |\mathbf{R}_j - \mathbf{R}_i| < r_c\}$$

- ε_i is a learned scalar function of the local environment within cutoff r_c
- Total cost is $\mathcal{O}(N)$ per energy evaluation

Consequences

- Naturally size-extensive: doubling the box doubles E
- Long-range physics (Coulomb, dispersion) must be added separately
- Equivariant message-passing networks effectively *learn* an enlarged r_c through repeated message rounds

Required symmetries

For any rotation $R \in O(3)$, translation $\mathbf{t}$, and same-species permutation π :

$$E(\{R\mathbf{R}_i + \mathbf{t}\}) = E(\{\mathbf{R}_i\}), \quad E(\{\mathbf{R}_{\pi(i)}\}) = E(\{\mathbf{R}_i\})$$

Two ways to enforce symmetry in an ML model:

Invariant features

Descriptors $\mathbf{d}_i$ already $O(3)$ -invariant; apply any NN on top.

- Easy to use, well-suited for energies
- Bottleneck: angular / directional info hard to recover

Equivariant features

Internal reps *transform* under $O(3)$ as spherical tensors Y_ℓ .

- Scalars, vectors, tensors coexist
- Forces & stresses arise naturally; higher body-order

Energies are scalars (invariant), forces are vectors, stresses are rank-2 tensors. Equivariant models predict all three consistently.

Forces, stresses, and training data

Energies, forces, and stresses come from the *same* function:

$$\mathbf{F}_i = -\nabla_{\mathbf{R}_i} E, \quad \sigma_{\alpha\beta} = -\frac{1}{V} \partial E / \partial \epsilon_{\alpha\beta}$$

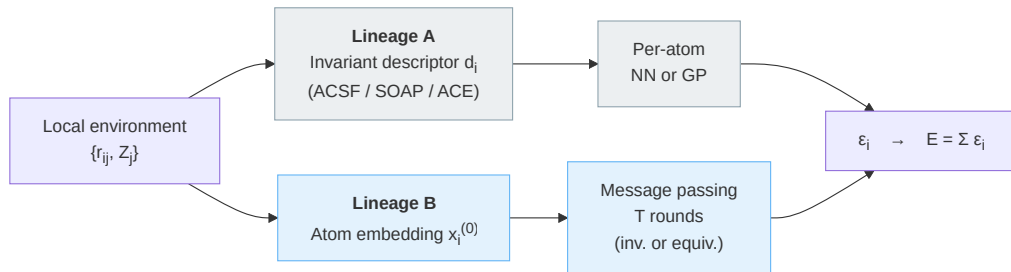
So for an ML model parametrized by θ :

$$\mathcal{L}(\theta) = \frac{1}{N_{\text{cfg}}} \sum_c \left[w_E \Delta E_c^2 + w_F \sum_i \|\Delta \mathbf{F}_{i,c}\|^2 + w_\sigma \|\Delta \boldsymbol{\sigma}_c\|^2 \right]$$

with $\Delta X_c = X_c^{\text{ML}} - X_c^{\text{DFT}}$.

- Forces give $3N$ labels per configuration, far richer than the single energy $\rightarrow$ almost always included in training
- Computed via **autodiff** through the ML model
- Modern datasets: MD17, ANI-1x, Materials Project, OC20/OC22, Alexandria, MPtrj — millions of DFT-labeled structures

Two architectural lineages



Lineage A — fixed descriptor $\rightarrow$ regressor.
Examples: Behler–Parrinello, GAP, ANI, DeepMD, ACE.

Lineage B — atoms as graph nodes, *learnable* messages.
Examples: SchNet, DimeNet, NequIP, MACE, Allegro.

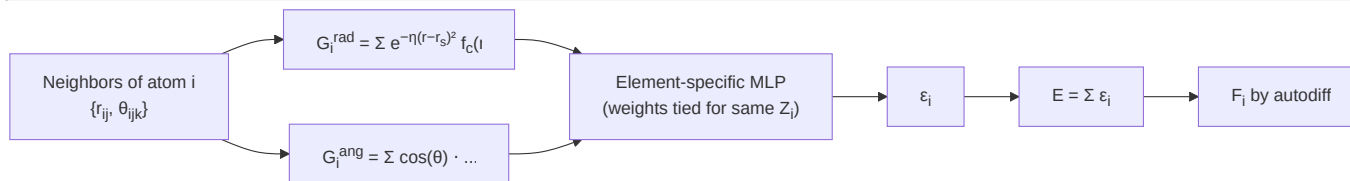
Equivariant message passing has been the dominant paradigm since 2022.

Part 3

Paper tour

Behler & Parrinello (2007)

"Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces" — Phys. Rev. Lett. **98**, 146401 (2007)



The recipe that defined the field

- Atomic energy decomposition $E = \sum_i \epsilon_i$
- Fixed-length, $O(3)$ -invariant *atom-centered symmetry functions* (ACSFs)
- One small MLP per element, weights shared over equivalent atoms
- First demonstration that NN potentials reach DFT accuracy on a dense dataset (Si)
- Limitations that seeded later work: ACSFs scale poorly with N_{species} , no formal completeness, hand-tuned η, r_s

GAP and SOAP (Bartók 2010, 2013)

Bartók, Payne, Kondor, Csányi, *Phys. Rev. Lett.* **104**, 136403 (2010) — **GAP**

Bartók, Kondor, Csányi, *Phys. Rev. B* **87**, 184115 (2013) — **SOAP**

GAP — Gaussian Approximation Potentials

Replace the per-atom NN with a **Gaussian Process** regressor.

- Probabilistic: predicts mean *and* uncertainty
- Smooth, data-efficient at small dataset sizes
- Cost grows with training set $\Rightarrow$ sparse GPs

SOAP — Smooth Overlap of Atomic Positions

A principled, $O(3)$ -invariant descriptor.

- Smooth Gaussian neighbor density, expanded in $R_n Y_{\ell m}$
- **Power spectrum** $p_i^{nn'\ell} = \sum_m c_{ilm}^n (c_{ilm}^{n'})^*$ is invariant
- Provably *complete* up to 3-body — basis for ACE

ANI (2017) and DeepMD (2018)

Smith, Isayev, Roitberg, *Chem. Sci.* **8**, 3192 (2017) — **ANI-1**

Zhang, Han, Wang, Saidi, Car, E, *Phys. Rev. Lett.* **120**, 143001 (2018) — **DeepMD / DPMD**

ANI-1

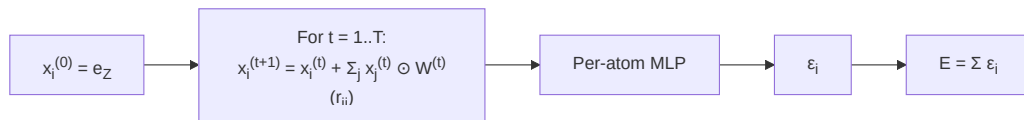
- Behler–Parrinello recipe at organic-chemistry scale
- Modified ACSFs (radial + angular grids)
- Trained on ~22M small-molecule configurations (ANI-1x dataset)
- First MLP that *transfers* across millions of organic molecules
- Established **active learning** as a standard tool to grow the training set

DeepMD

- *Drop the hand-crafted descriptor*. learn it end-to-end
- "Embedding net" maps each neighbor pair to a learned vector; symmetric pooling makes it $O(3)$ -invariant
- Per-atom "fitting net" outputs ϵ_i
- Open-source (DeePMD-kit), strong adoption in materials science
- **Gordon Bell 2020**: 100M-atom MD with ab initio accuracy

SchNet (2017)

Schütt, Kindermans, Sauceda, Chmiela, Tkatchenko, Müller — NeurIPS 2017 / *J. Chem. Phys.* **148**, 241722 (2018)



The first widely-used graph NN potential

- Continuous radial filters $W(r_{ij})$ — no distance discretization
- Repeated rounds of message passing $\Rightarrow$ effective receptive field grows with depth
- **$O(3)$ -invariant** (only uses r_{ij} , not directions)
- Showed graph message passing could match descriptor methods on MD17
- Limitation: ignoring angles $\Rightarrow$ insensitive to chirality, weak on dense angular features

DimeNet & DimeNet++ (2020)

Klicpera, Groß, Günnemann — ICLR 2020 / *NeurIPS* 2020

Adding directional information without breaking invariance.

- Messages are now indexed by *triplets* (j, i, k) , encoding bond angles θ_{ijk}
- Uses spherical Bessel + spherical harmonic basis for (r_{ij}, θ_{ijk})
- Stays $O(3)$ -invariant but can resolve angular environments

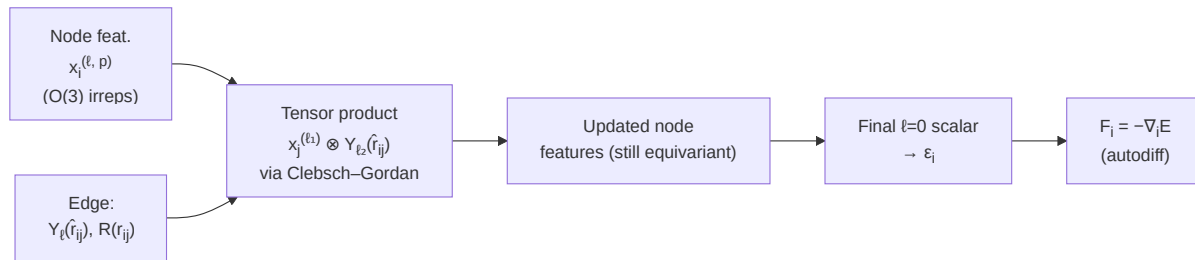
$$m_{ji} \rightarrow m'_{ji} = f(m_{ji}, \{m_{kj} : k \in \mathcal{N}(j) \setminus i\}, r_{ij}, \theta_{kji})$$

Why it matters

- First to clearly outperform SchNet on QM9 by including angular information
- Bridge between purely invariant message passing and the equivariant wave that came next
- Predecessor of **GemNet** (2021), which adds dihedrals — full geometric info

NequIP (2022)

Batzner, Musaelian, Sun, Geiger, Mailoa, Kornbluth, Molinari, Smidt, Kozinsky — *Nat. Commun.* **13**, 2453 (2022)



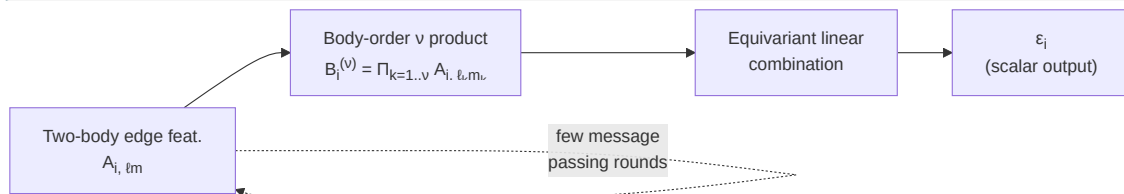
$E(3)$ -equivariant message passing

- Node features are *lists of spherical tensors* $\mathbf{x}_i^{(\ell, p)}$, not just scalars
- Messages preserve equivariance via Clebsch–Gordan products with edge Y_ℓ
- 1–2 orders of magnitude **more data efficient** than invariant models on MD17 / rMD17 / water
- Founded a family: **Allegro, MACE, SEGNN, EquiformerV2**

The "NequIP moment" is when symmetry-aware deep learning became the default for ML potentials.

MACE (2022)

Batatia, Kovács, Simm, Ortner, Csányi — *NeurIPS 2022*



Higher body-order in a single layer

- Combines the **Atomic Cluster Expansion** (ACE; Drautz 2019) with $E(3)$ -equivariant message passing
- One MACE layer encodes 3-body and 4-body correlations; SchNet/NequIP need many layers

$$E_i = \sum_{\nu=1}^{\nu_{\max}} \sum_{lm} c_{\nu lm}^{(i)} \prod_{k=1}^{\nu} A_{i, \ell_k m_k}$$

- Few layers $\Rightarrow$ faster per step *and* longer effective receptive field
- State-of-the-art on rMD17, SPICE, water, organic crystals; backbone of **MACE-MP-0** (2024)

Allegro (2023)

Musaelian, Batzner, Johansson, Sun, Owen, Kornbluth, Kozinsky — *Nat. Commun.* **14**, 579 (2023)

Strictly local equivariant model, designed for MD scale.

- Equivariant tensor-product features, **no message passing between atoms** — every atomic energy depends only on neighbors within r_c
- Pure two-body neighbor list $\Rightarrow$ embarrassingly parallel, scales to *millions* of atoms on GPUs
- Multiple "layers" act on edge features but never propagate beyond direct neighbors

Why it matters

- Demonstrates that strict locality + rich equivariant features can match graph models in accuracy
- Removes the long-range data-dependency that hurts MD parallelization in deep MPNNs
- A practical answer to: "*how do we make equivariant MLPs run at LAMMPS scale?*"

Comparison cheat-sheet

Model	Year	Symmetry	Body order / layer	Trick
Behler–Parrinello	2007	invariant (ACSF)	2 + 3 (hand-crafted)	per-element NN
GAP + SOAP	2010/13	invariant (SOAP)	up to 3 (complete)	Gaussian process
ANI	2017	invariant (mod. ACSF)	2 + 3	organic dataset, active learning
DeepMD	2018	invariant (learned)	2 (learned embedding)	end-to-end descriptor
SchNet	2017	invariant	2 (radial only)	continuous filters
DimeNet++	2020	invariant	2 + 3 (angles)	triplet messages
NequIP	2022	$E(3)$ -equivariant	2 per layer, deep	Clebsch–Gordan tensor products
MACE	2022	$E(3)$ -equivariant	up to 4 (ACE)	high body-order in one shot
Allegro	2023	$E(3)$ -equivariant	2 (strictly local)	scalable MD

Reading order: Behler–Parrinello → SchNet → NequIP → MACE → Allegro.

Part 4

Foundation models, applications, open problems

Universal / foundation MLPs

The 2022–2024 wave: trained on the *full* periodic table, millions of DFT structures.

Model	Year	Backbone	Training data
M3GNet (Chen & Ong)	2022	invariant graph NN	Materials Project relaxations
CHGNet (Deng <i>et al.</i>)	2023	invariant + magnetic moments	MPtrj (1.5M structures)
MACE-MP-0 (Batatia <i>et al.</i>)	2024	MACE	MPtrj
MatterSim (Microsoft)	2024	M3GNet-style	~17M structures, T/P aware
ORB (Orbital Materials)	2024	graph transformer	MPtrj + Alexandria
SevenNet / Equiformer-V2	2024	equivariant transformer	MPtrj, OC20

Single checkpoint, *zero-shot* useful across the periodic table; compute shifts to fine-tuning on top of a pretrained model. ML potentials are starting to look like NLP: pretrain broad, fine-tune narrow. Benchmarks: **Matbench-Discovery**, MD17/SPICE.

Applications

What MLPs let us actually do (with DFT-level fidelity, at 10^4 – 10^8 atoms):

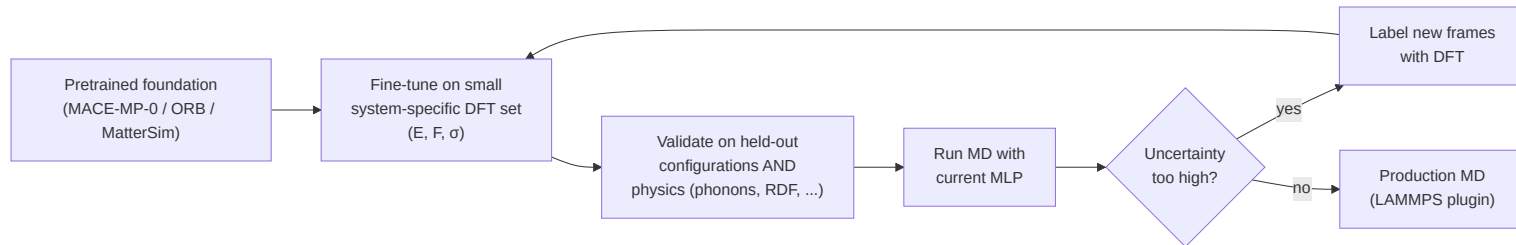
- **Phase transitions and thermodynamics** — water phases, Si melting, Ga, alloy order/disorder
- **Defect dynamics** — vacancy diffusion, dislocation glide, GB migration
- **Reactive chemistry** — proton transfer, catalysis on surfaces, electrochemical interfaces (with ML + electrostatics)
- **Soft matter and biomolecules** — peptide conformations, protein folding pathways at near-DFT accuracy
- **High-throughput materials discovery** — screen candidate compounds for stability and properties before running DFT
- **Replacing parts of DFT loops** — Δ -learning, ML-accelerated NEB, ML-augmented free-energy sampling
- **Driver for differentiable simulation** — fully autodiff'd MD pipelines (JAX-MD + MACE/Allegro)

A useful slogan: *"DFT for the dataset, ML for the dynamics."*

Open problems

- **Long-range interactions** — locality assumption breaks for ionic / polar systems and Coulomb. Active research: 4G-HDNNP (charge equilibration NN), latent Ewald, SCFNN, long-range message passing.
- **Charge & spin states** — most MLPs assume a single charge / spin sector. Magnetic materials need spin-aware architectures (CHGNet's magnetic moments are an early step).
- **Excited states & non-adiabatic dynamics** — beyond Born–Oppenheimer; emerging diabatic-state ML approaches.
- **Uncertainty quantification & active learning** — ensembling, GP last layer, deep kernel — still an art for choosing what to label next.
- **Out-of-distribution behavior** — extrapolation to new chemistries, extreme T/P, never-seen reactions; foundation-model scaling helps but doesn't solve it.
- **Cost vs. accuracy at scale** — MD throughput per GPU still trails best classical FFs by orders of magnitude. Allegro / kernel reductions / mixed precision are pushing this.

Practical playbook (for your term project)



1. **Start from a pretrained foundation model** (MACE-MP-0, ORB, MatterSim) — fine-tune on a few hundred DFT-labeled configurations of *your* system.
2. **Use an existing framework** — MACE, NequIP, Allegro all have clean PyTorch/JAX implementations and clear configs.
3. **Validate on held-out *configurations* AND held-out *physics*** — RMSE is necessary but not sufficient; check phonons, elastic constants, RDFs vs. DFT MD.
4. **Active learning loop** — run MD, flag high-uncertainty frames (ensemble disagreement / GP variance), label with DFT, retrain.
5. **Production MD** — LAMMPS plugins exist for all major architectures. Profile carefully; communication is often the bottleneck.

Summary

- **The setting** — learn $E(\{\mathbf{R}_i, Z_i\})$ to bridge the cost/accuracy gap between DFT and classical force fields.
- **The universal recipe** — atomic energy decomposition + symmetry-respecting representation of local environments.
- **The two lineages** — invariant descriptors + per-atom NN (Behler–Parrinello → DeepMD), and equivariant message passing on graphs (SchNet → NequIP → MACE → Allegro).
- **Today** — foundation-scale pretrained models (MACE-MP-0, MatterSim, ORB) trained on millions of DFT structures, fine-tuned per problem.
- **Frontiers** — long-range physics, charge/spin states, true UQ, and bringing throughput closer to classical FFs.

Reading list

Reviews

- Unke *et al.*, *Chem. Rev.* **121**, 10142 (2021) — *Machine learning force fields*
- Behler, *Chem. Rev.* **121**, 10037 (2021) — *Four generations of HDNNPs*
- Friederich *et al.*, *Nat. Mater.* **20**, 750 (2021)

Foundational papers

- Behler & Parrinello, *PRL* **98**, 146401 (2007)
- Bartók *et al.*, *PRL* **104**, 136403 (2010); *PRB* **87**, 184115 (2013)
- Smith *et al.*, *Chem. Sci.* **8**, 3192 (2017); Zhang *et al.*, *PRL* **120**, 143001 (2018)
- Schütt *et al.*, *J. Chem. Phys.* **148**, 241722 (2018); Klicpera *et al.*, *ICLR* 2020
- Batzner *et al.*, *Nat. Commun.* **13**, 2453 (2022) — NequIP
- Batatia *et al.*, *NeurIPS* 2022 (MACE); arXiv:2401.00096 (2024) (MACE-MP-0)
- Musaelian *et al.*, *Nat. Commun.* **14**, 579 (2023) — Allegro

Thank you

Questions?

- Slides and code: see course website
- Office hours: Mon, Tue, Thu 10:00–12:00 (F303C)

Next lecture (Lecture 12): Equivariant Neural Networks — the math behind NequIP / MACE / Allegro.